LAB SESSION 5

Programming Exercise:

Question 1: Find the sum of all elements in a 2D array.

C: > C++ PROGRAMS > tut1.cpp > main()

```cpp
#include <iostream>
using namespace std;

const int ROW = 3;
const int COL = 3;

// Function to find the sum of all elements in a 2D array
int sumMatrix(int matrix[ROW][COL])
{
    int sum = 0;

    for (int i = 0; i < ROW; i++)
    {
        for (int j = 0; j < COL; j++)
        {
            sum += matrix[i][j]; // Add each element to sum
        }
    }

    return sum;
}
int main()
{
    int matrix[ROW][COL] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}};

    int total = sumMatrix(matrix);

    cout << "Sum of all elements = " << total << endl;

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Sum of all elements = 45
PS C:\C++ PROGRAMS>
```

Question2: Calculate the transpose of a given matrix.

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Original Matrix:
1 2 3
4 5 6
7 8 9

Transpose Matrix:
1 4 7
2 5 8
3 6 9
PS C:\C++ PROGRAMS>
```

```cpp
#include <iostream>
using namespace std;

const int ROW = 3;
const int COL = 3;

// Function to calculate transpose of a matrix
void transposeMatrix(int matrix[ROW][COL], int transpose[COL][ROW])
{   for (int i = 0; i < ROW; i++) {
        for (int j = 0; j < COL; j++) {
            transpose[j][i] = matrix[i][j];  // Swap row and column
        }
    }
}

int main() {
    int matrix[ROW][COL] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int transpose[COL][ROW];

    // Calculate transpose
    transposeMatrix(matrix, transpose);

    // Display original matrix
    cout << "Original Matrix:" << endl;
    for (int i = 0; i < ROW; i++) {
        for (int j = 0; j < COL; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }

    // Display transpose matrix
    cout << "\nTranspose Matrix:" << endl;
    for (int i = 0; i < COL; i++) {
        for (int j = 0; j < ROW; j++) {
            cout << transpose[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Question3: Check if a matrix is symmetric or not.

```cpp
#include <iostream>
using namespace std;

const int SIZE = 3;  // For a 3x3 matrix

// Function to check if a matrix is symmetric
bool isSymmetric(int matrix[SIZE][SIZE]) {
    for (int i = 0; i < SIZE; i++) {
        for (int j = 0; j < SIZE; j++) {
            if (matrix[i][j] != matrix[j][i]) {
                return false;  // Not equal to its
transpose    }
        }
    }
    return true;  // All elements matched
}

int main() {
    int matrix[SIZE][SIZE] = {
        {1, 2, 3},
        {2, 5, 6},
        {3, 6, 9}
    };

    // Display matrix
    cout << "Matrix:\n";
    for (int i = 0; i < SIZE; i++) {
        for (int j = 0; j < SIZE; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }

    // Check for symmetry
    if (isSymmetric(matrix))
        cout << "\nThe matrix is symmetric." << endl;
    else
        cout << "\nThe matrix is not symmetric." << endl;

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Matrix:
1 2 3
2 5 6
3 6 9

The matrix is symmetric.
PS C:\C++ PROGRAMS>
```

Question4: Multiply two matrices and print the result.

```cpp
#include <iostream>
using namespace std;

const int ROW1 = 2; // Rows of first matrix
const int COL1 = 3; // Columns of first matrix
const int ROW2 = 3; // Rows of second matrix (must be equal to COL1)
const int COL2 = 2; // Columns of second matrix

// Function to multiply two matrices
void multiplyMatrices(int matrix1[ROW1][COL1], int matrix2[ROW2][COL2], int result[ROW1][COL2])
{   for (int i = 0; i < ROW1; i++) {
        for (int j = 0; j < COL2; j++) {
            result[i][j] = 0; // Initialize result cell
            for (int k = 0; k < COL1; k++) {
                result[i][j] += matrix1[i][k] * matrix2[k][j];
            }
        }
    }
}

// Function to print matrix
void printMatrix(int matrix[][COL2], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }
}

int main() {
    int matrix1[ROW1][COL1] = {
        {1, 2, 3},
        {4, 5, 6}
    };

    int matrix2[ROW2][COL2] = {
        {7, 8},
        {9, 10},
        {11, 12}
    };

    int result[ROW1][COL2]; // Resultant matrix

    // Multiply the matrices
    multiplyMatrices(matrix1, matrix2, result);

    // Print the result
    cout << "Resultant Matrix after Multiplication:\n";
    printMatrix(result, ROW1, COL2);

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Resultant Matrix after Multiplication:
58 64
139 154
PS C:\C++ PROGRAMS>
```

Question5: Find the largest element in each row of a matrix.

```cpp
#include <iostream>
using namespace std;

const int ROW = 3;  // Number of rows in the matrix
const int COL = 3;  // Number of columns in the matrix

// Function to find the largest element in each row
void findLargestInRow(int matrix[ROW][COL], int maxInRow[ROW]) {
    for (int i = 0; i < ROW; i++) {
        int max = matrix[i][0];  // Start with the first element of the row
        for (int j = 1; j < COL; j++) {
            if (matrix[i][j] > max) {
                max = matrix[i][j];  // Update max if a larger element is
found       }
        }
        maxInRow[i] = max;  // Store the largest element in the result array
    }
}

int main() {
    int matrix[ROW][COL] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int maxInRow[ROW];  // Array to store the largest element of each row

    // Find largest element in each row
    findLargestInRow(matrix, maxInRow);

    // Print the largest element in each row
    cout << "Largest elements in each row:" << endl;
    for (int i = 0; i < ROW; i++) {
        cout << "Row " << i + 1 << ": " << maxInRow[i] << endl;
    }

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Largest elements in each row:
Row 1: 3
Row 2: 6
Row 3: 9
PS C:\C++ PROGRAMS>
```

Question 6: Reverse the elements of a given vector without using the reverse() function.

```cpp
#include <iostream>
#include <vector>
using namespace std;

// Function to reverse a vector
void reverseVector(vector<int>& vec) {
    int start = 0;          // Start index
    int end = vec.size() - 1;  // End index

    // Swap elements from start and end until they
meetwhile (start < end) {
        swap(vec[start], vec[end]);
        start++;
        end--;
    }
}

int main() {
    // Example input vector
    vector<int> vec = {1, 2, 3, 4, 5};

    // Reverse the vector
    reverseVector(vec);

    // Output the reversed vector
    cout << "Reversed vector: {";
    for (size_t i = 0; i < vec.size(); i++) {
        cout << vec[i];
        if (i != vec.size() - 1) {
            cout << ", ";
        }
    }
    cout << "}" << endl;

    return 0;
}
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($
?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Reversed vector: {5, 4, 3, 2, 1}
PS C:\C++ PROGRAMS>
```